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Predictive models for the implementation of targeted reproductive management in multiparous cows on automatic milking systems

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ABSTRACT

Targeted reproductive management (TRM) aims to improve the fertility efficiency of the dairy herd by applying group-level management strategies based on expected reproductive performance. Key to the utility of TRM is the accuracy with which an animal's reproductive performance can be predicted. Automatic milking systems (AMS) allow for the collection of data relating to milk quantity, quality, and robot visit behavior throughout the transition period. In addition to this, auxiliary data sources, such as rumination and activity monitors, as well as historical cow-level data, are often readily available. The utility of this data for the prediction of fertility has not been previously explored. The first objective of this study was to assess the accuracy with which the likelihood of expression of estrus between 22 and 65 DIM and conception to first insemination between 22 and 80 DIM could be predicted using data collected by AMS from 1 to 21 DIM. Our second objective was to assess the change in model performance following the addition of 2 auxiliary data sources. Using data derived solely from the AMS (RBT dataset) a binary random forest classification model was constructed for both outcomes of interest. The performance of these models was compared with models constructed using AMS data in conjunction with 2 auxiliary sources (RBT+ dataset). Expression of estrus was classified with an area under the receiver operator curve (AUC-ROC) of 0.6 and 0.65, conception to first insemination with an AUC-ROC of 0.56 and 0.62 for the RBT and RBT+ datasets, respectively. No statistically significant improvement in classification accuracy was achieved by the addition of auxiliary data sources. This is the first study to report the utility of data collected by AMS for the prediction of reproductive performance. Though the performance described is comparable with previously reported models, their utility for the implementation of TRM is limited by poor classification accuracy within key subgroups. Of note within this study is the failure of the addition of auxiliary data sources to increase the accuracy of prediction over models built using AMS data alone. We discuss the advantages and limitations the integration of additional data sources imposes on model training and deployment and suggest alternative methods to improve performance while preserving model parsimony.

Key words: machine learning, transition cow, automatic milking, precision technology, targeted reproductive management

INTRODUCTION

Targeted reproductive management (**TRM**) aims to improve the fertility efficiency of the dairy herd by applying bespoke group-level management strategies based on expected reproductive performance (Giordano et al., 2022). This is achieved through a 3-step process. First, the development of models for the prediction of reproductive performance; second, the classification of animals into groups based on these predictions; and finally, the implementation of targeted management strategies designed to optimize reproduction. Examples of its application include the targeted use of reproductive hormones in animals at reduced risk of estrus expression in early lactation (Rial et al., 2022; Gonzalez et al., 2023) and the preferential use of sexed semen in animals deemed highly likely to conceive to AI (Berry, 2021). Models capable of predicting performance in these areas would be of value in the implementation of TRM.

The influence of the transition period on the subsequent fertility of the dairy cow is well documented (Walsh et al., 2011; Chapinal et al., 2012). The success with which the cow adapts to the stressors of early lactation has a demonstrated effect on expression of estrus (Vercouteren et al., 2015; Banuelos and Stevenson 2021) and risk of conception (Elkjær et al., 2013; Caixeta et

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-24. Nonstandard abbreviations are available in the Notes.

al., 2017; Mohtashamipour et al., 2020). As the adoption of sensor technology on commercial dairy farms continues to increase, so too does the opportunity to harvest data reflective of the cow's physiological status during transition. These data may prove to have utility for the development of the predictive models that form the basis of TRM.

Dairy farms employing automatic milking systems (**AMS**) offer a unique opportunity to collate transition cow data for use in such models. A wide range of variables relating to milk quantity, quality, concentrate intake, and robot visit behavior are automatically generated from sensors incorporated into the milking robot. In addition to this, auxiliary data sources, such as neck mounted accelerometers as well as historical cow-level data, are often readily available. The integration of a wide range of data sources in model development has the potential to improve the accuracy of predictions by providing a more detailed representation of each animal. However, an increased quantity of data does not guarantee improved model performance (Berisha et al., 2021). Furthermore, as the number of data sources used increases, the ease with which the model can be deployed in a commercial setting is reduced (Leff and Lim, 2021). The development of predictive models should therefore aim to balance model accuracy with parsimony by minimizing the number of data sources used.

The objective of this study was first, to assess the accuracy with which the likelihood of expression of estrus and conception to first insemination could be predicted using data collected by AMS from 1 to 21 DIM. A second objective was to assess the change in model performance following the addition of 2 auxiliary data sources.

MATERIALS AND METHODS

This study was conducted following ethical approval from the School of Veterinary Medicine and Science, University of Nottingham, Committee for Animal Research and Ethics (Reference No. 3404 210708). All analysis was carried out using R statistical software (R Core Team, 2020). Thirty-six commercial AMS herds from the United Kingdom and Republic of Ireland were enrolled on a voluntary basis in this retrospective cohort study. Criteria for inclusion was the use of a Lely Astronaut Milking Robot (Lely International N.V.) under free flow traffic conditions (Munksgaard et al., 2011), in conjunction with rumination and activity monitoring technology (Lely Qwes-HR collars, Lely International N.V.). Participating herd data from January 2017 to August 2023 was available via Lely's Business Integration Application Programming Interface. Data relating to milk quantity and quality, the frequency of cow-robot interactions, rumination and activity, as well as reproductive management records were accessed. No information pertaining to the use of estrus synchronization or fixed-time insemination was available for animals within this dataset.

Data Preparation

Two outcomes of interest were selected for investigation. The first was an expression of estrus or insemination event (**EOI**). An animal was classed as EOI+, where an estrus or insemination event was recorded between d 22 and 65 postcalving. An estrus event was defined as 3 consecutive 2-h periods of increased activity compared with each animal's predetermined baseline as detected by a neck mounted activity monitor. Records relating to insemination events were obtained from the on-farm management system. The second outcome of interest was conception to first insemination (**CFI**). Animals were classed as CFI+ where they received their first and only insemination between d 22 and 80 postcalving and were subsequently recorded as pregnant on the farm management system.

For each outcome of interest, 2 datasets were constructed. The first was composed solely of data provided by the AMS (the **RBT** dataset; Supplemental Table S1, see Notes). The second was composed of AMS data in conjunction with 2 auxiliary data sources (the **RBT+** dataset (Supplemental Table S2, see Notes). The aim of this study was to compare the accuracy with which EOI and CFI could be predicted using their respective RBT and RBT+ datasets. To allow the comparison of RBT and RBT+ using identical subjects, cow-lactations were assessed for data missingness across all 3 data sources. Only cow-lactations with sufficient data across all 3 were retained. Following feature engineering, fertility data for all retained cow-lactations were assessed and an outcome for EOI and CFI assigned. Those without sufficient data to allow determination of their fertility performance were removed. For each of the outcomes EOI and CFI, an RBT and RBT+ dataset were brought forward for model development. This data preparation process is described in detail below and displayed in Figure 1.

Feature Engineering: AMS Data

The day of calving was designated day zero, data collected by the AMS from d 1 to 21 was used to engineer 10 features for inclusion in the RBT dataset. Milk quantity was assessed as mean milk yield (kg; the mean of daily milk yields over d 1–21), and mean delta yield (%; the mean change in daily milk yield from consecutive days, expressed as a percentage of the first). For example, mean delta yield for each cow-lactation was calculated as the change in yield from d 1 to 2, expressed as a per-

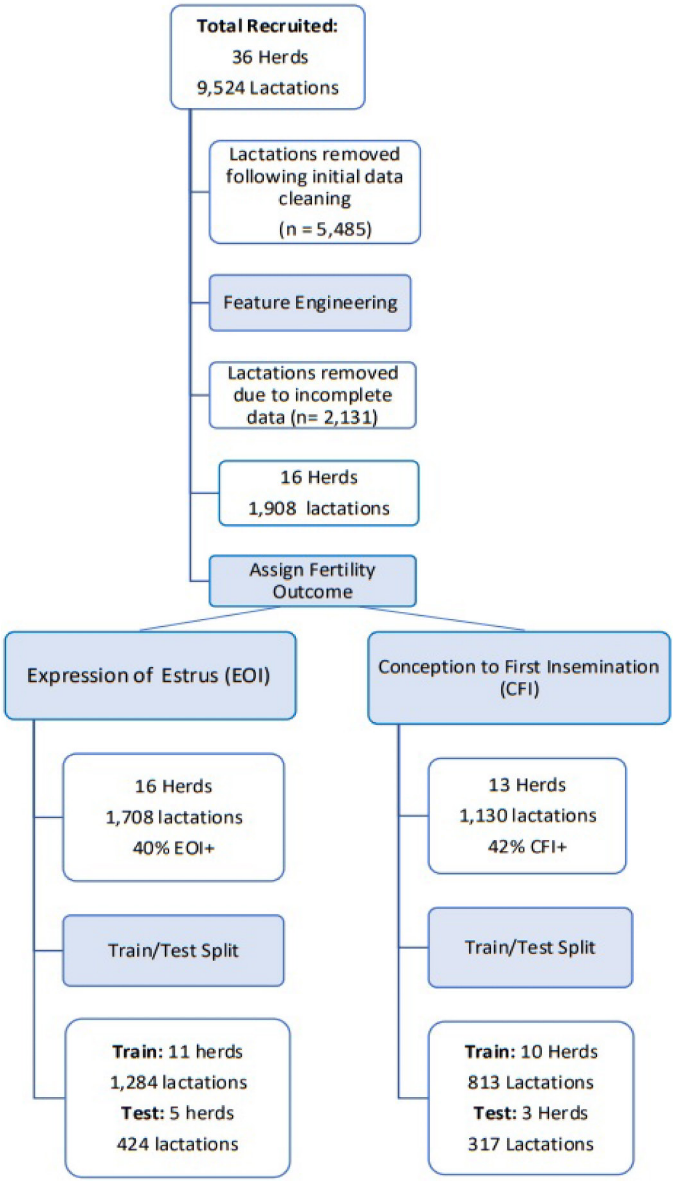

Figure 1. Workflow for data cleaning, feature engineering, and preparation of the final data sets used for model construction and external validation.

centage of d 1, averaged with, the change in yield from d 2 to 3, expressed as a percentage of d 2 yield and so on. Milk quality was assessed using milk conductivity and constituent data. Two measures of milk conductivity were employed, mean conductivity (arbitrary unit, **AU**); the mean udder level conductivity recorded over d 1 to 21 and conductivity alert; the total number of instances where quarter level conductivity exceeded 80 units. Milk fat and protein indications, as recorded once daily by Lely's MQC-C (Fadul-Pacheco et al., 2018) were used as mean fat and mean protein; the mean of recorded values across d 1 to 21 for their respective constituent. The ratio of mean fat to mean protein was subsequently calculated. Mean concentrate intake (kg) was calculated as the mean kilograms of concentrate dispensed to each animal by the robot per day. Robot visit behavior parameters consisted of milking visits and milking refusals (where milking permission is denied due to an animal re-presenting a short time after a previous milking visit). These were averaged over d 1 to 21 and reported as mean milkings and mean refusals. To be retained in the final dataset, measurements for each of these 10 parameters were required for at least 16 of the first 21 d postcalving. Where these were not available, the entire cow-lactation was removed.

Feature Engineering: Auxiliary Data Sources

Rumination and activity data derived from neck mounted accelerometers were used to engineer 7 variables (Schirmann et al., 2009). Animals that failed to record complete rumination and activity values for at least 16 of the 21 d analyzed were removed from the analysis. Engineered variables were mean rumination (AU; the mean daily rumination recorded), mean delta rumination (the mean change in daily rumination), and mean variation rumination (the variance in mean rumination across the 21-d period). These 3 metrics were also calculated for data relating to daily activity. In addition, the sum of heats recorded by the neck mounted accelerometers over d 1 to 21 postcalving was assessed as the transition heat count.

Cow-level historical data as recorded by the on-farm management system were used to engineer 7 variables. For each cow-lactation, records describing milk quantity, quality, and fertility performance for the prior lactation were assessed. Milk yield in the prior lactation was assessed as daily milk yield averaged across the entire lactation (**PV-mean milk yield**). Mean milk yield for the first 30 d of the previous lactation, mean milk yield for the last 10 d before drying off (**PV-yield at dry**), and conductivity of milk in the 10 d before dry-off were also examined. The DIM at dry-off (**PV-DO**) and the number of days dry were assessed. Finally, DIM at conception in the prior lactation was addressed as **PV-DC**, and age at first calving as **AFC**. The necessity of a prior lactation for the assessment of these variables resulted in the removal of all primiparous animals from the dataset. Before modeling, all numeric variables were centered by subtracting the column means of each variable from their corresponding columns and scaled by dividing each variable by their SD.

Following feature engineering, an EOI and CFI status was assigned to retained cow-lactations. For each outcome of interest, animals for which no definitive status could be established (i.e., EOI+ or EOI- and CFI+ or

CFI-) were removed. In the case of EOI, animals without an estrus or insemination event which also failed to log complete activity records for d 22 to 65 were removed. For example, any animal for which no estrus or insemination was recorded but for which a complete activity record was not available was removed from analysis, as we could not rule out the occurrence of estrus on the days for which activity records were absent. In the case of CFI, animals which did not receive a first insemination between 22 and 80 DIM and those inseminated after the end of our observation window were removed. Following assignment of EOI and CFI status, herds with less than 50 cow-lactations available for each outcome were removed. Thereafter, the datasets for both EOI and CFI were split into RBT and RBT+, comprising 10 and 25 variables, respectively. The former was composed solely of data provided by the AMS, the latter comprised AMS data in conjunction with both auxiliary data sources. These 4 datasets were brought forward for model construction. The EOI and CFI datasets contained 1,708 multiparous lactation from 16 herds, and 1,130 multiparous lactations from 13 herds, respectively (Figure 1).

Model Construction and Evaluation

To assess the accuracy with which the RBT dataset could predict the likelihood of both EOI and CFI, predictive models for each outcome of interest were constructed and externally validated using the procedure described below. The dataset was split randomly by herd into a train and test dataset. All features within the training dataset were offered to a random forest recursive feature elimination (**RFE**) model. The optimal subset of variables was assessed by the effect of their inclusion on the area under the receiver operator curve (**AUC-ROC**; Kuhn and Johnson, 2013). Retained variables were selected based on examination of RFE plots generated by the VarImp function in the CARET package (Kuhn, 2008) with the goal of balancing parsimony and performance. Those retained were brought forward to a final random forest model (Breiman, 2001) trained to maximize AUC-ROC over 5-fold cross validation repeated 10 times. The scaled variable importance for features retained in the final model was assessed using the VarImp function. This model building procedure was repeated using the RBT+ dataset.

The final predictive models EOI-RBT, EOI-RBT+, CFI-RBT, and CFI-RBT+ were evaluated by their predictive performance using the test dataset. Model performance was assessed using classification accuracy, sensitivity, specificity, positive predictive value (**PPV**), negative predictive value (**NPV**), and AUC-ROC. Within each outcome of interest, the classification accuracy for the RBT and RBT+ models were compared using McNemar's test (Dietterich, 1997). The null hypothesis was that no difference in classification accuracy exists between the RBT and RBT+ models, a significance level of 0.1 was employed. To investigate the accuracy with which these models might be applied on farm, test set predictions were ranked by probability within their respective farms and segregated into quartiles (**Q**). Those in Q1 represented animals least likely, within their respective herds, to record a positive outcome (an estrus or insemination event in the case of EOI, conception to first insemination in the case of CFI). Those in Q4 represented the most likely. Of particular interest in our study were Q1 and Q4 as they represented key groups that may be selected for TRM. For example, animals classified as Q1 for EOI may be selected for blanket hormone treatment, whereas those in Q4 may be observed for natural estrus without hormonal intervention. Similarly, those in Q4 for CFI may be selected for insemination with sexed semen, whereas those in Q1 may receive beef semen.

RESULTS

Complete or partial records relating to 9,524 lactations from 36 recruited herds were assessed. Initial data cleaning involved the removal of primiparous lactations, those that failed to reach 21 d postcalving and those that failed to record any corresponding milk production, rumination, or activity data. A total of 4,039 multiparous lactations from 34 herds were brought forward for feature engineering. Incomplete milk quality, rumination, or activity data led to the removal of 2,131 lactations. From the remaining 1,908 lactations, those without corresponding fertility records and those originating from herds with less than 50 cow-lactations available for analysis were removed. Descriptive statistics for herds retained in the final datasets are presented in Table 1. A total of 1,708 cow-lactations from 16 herds formed the final EOI dataset, of which 40% were classed as EOI+. The herd-level mean DIM at which an EOI event was recorded was 51, with a range of 45 to 55 DIM. The final CFI dataset was composed of 1,130 cow-lactations from 13 herds of which 42% were classed at CFI+. The herd-level mean DIM at first insemination was 61, ranging from 54 to 70 DIM. Descriptive statistics for the cow-lactations retained in the final EOI and CFI datasets are presented in Tables 2 and 3, respectively. Frequency distributions for EOI and CFI events are presented in Supplemental Figures S1 and S2 (see Notes), respectively.

Expression of Estrus and Insemination Events

The EOI dataset was divided into a training and test dataset of 1,284 lactations from 11 herds and 424 lactations from 5 herds, respectively. Following assessment

Table 1. General descriptive statistics for herds which contributed cow-lactations to EOI and CFI dataset

Variable	Total (range)
Total number of herds	16
Production system	
All year-round calving, fully housed	13
Spring calving, grazing	3
Geographical region	
England	9
Republic of Ireland	3
Northern Ireland	3
Wales	1
Mean no. of milking cows/herd	215 (57–419)
Mean no. of AMS units/herd	3 (2–7)
Mean 305-d yield, herd level (kg)	8,438 (5,216–12,825)
Mean percentage of animals inseminated by 80 DIM, herd level (%)	74 (46–95)

of the RFE plot (Supplemental Figure S3, see Notes), 5 variables, the combination of which resulted in the highest AUC-ROC were retained in the final EOI-RBT model. These were, mean milk yield, mean concentrate intake, mean protein, mean refusals, and mean conductivity. The scaled variable importance for each is presented in Table 4 and descriptive statistics in Supplemental Tables S1 and S2. Animals were classified with an AUC-ROC of 0.6 (Figure 2), sensitivity of 38%, specificity of 79%, PPV, NPV, and accuracy of 60% (Table 5).

In the final EOI-RBT+ model, 8 variables including at least one from each of the 3 data sources were retained following RFE (Supplemental Figure 4, see Notes); mean concentrate intake, mean variation activity, PV-mean milk yield, mean activity, mean milk yield, AFC, mean protein, and PV-DC. Scaled importance for all variables retained in the final model is presented in Table 6. Animals were classified with an AUC-ROC of 0.65, sensitivity of 15%, specificity of 91%, PPV of 59%, NPV of 57%, and accuracy of 57%.

When the accuracy of both models was compared, a P-value of 0.9 was obtained and the null hypothesis was accepted, indicating there was no significant difference in performance between models (Table 5). The accuracy with which EOI was predicted within farm level quartiles is presented in Table 7. Among animals in Q1, an accuracy of 61% and 72% was recorded for the RBT and RBT+ models, respectively. Animals in Q4 were classified with an accuracy of 60% by RBT and 48% by RBT+ models.

Table 2. Descriptive statistics for cow-lactations retained in the final EOI and EOI+ random forest models for the prediction of expression of estrus between DIM 22 and 65

Variable	Total (range)
No. of cow-lactations	1,708
No. of farms	16
Production system	
All year-round calving, fully housed	13
Spring calving, grazing	3
Mean no. of cow-lactations/herd	106 (52–352)
Mean peak yield,[1] herd level (kg)	54 (38–69)
Expression of estrus (EOI+)	
Total (%)	40
Parity 2 (%)	45
Parity 3+ (%)	36
Mean EOI + animals, herd level (%)	42 (20–73)
Mean DIM at EOI+ event, herd level (d)	51 (45–55)

[1]Calculated using maximum single-day yield recorded in the first 100 DIM.

Conception to First Insemination

The CFI dataset was divided into a training set of 813 lactations from 10 herds and test set of 317 lactations from 3 herds. Seven variables were retained in the CFI-RBT model following RFE (Supplemental Figure S5, see Notes); mean delta yield, mean concentrate intake, mean conductivity, mean milk yield, mean fat-to-protein ratio, mean protein, and mean milkings. Scaled variable importance are presented in Table 8, descriptive statistics are presented in Supplemental Tables S1 and S2. Animals that conceived at first insemination were classified with an AUC-ROC of 0.56 (Figure 3), sensitivity of 25%, specificity of 86%, a PPV of 53%, NPV of 64%, and accuracy of 62% (Table 5).

Eleven variables were retained in the final CFI-RBT+ model (Supplemental Figure S6, see Notes); mean conductivity, mean concentrate intake, mean delta yield, mean milkings, mean fat-to-protein ratio, mean protein, mean rumination, mean delta rumination, PV-DO, PV-yield at dry, and PV-mean milk yield (Table 9). Animals were classified with an AUC-ROC of 0.62, sensitivity of 25%, specificity of 80%, PPV of 44%, NPV of 63%, and accuracy of 59%. When the accuracy of RBT and RBT+ models were compared using McNemar's test, a P-value of 0.5 was obtained and the null hypothesis accepted (Table 5). The accuracy with which RBT and RBT+ models predicted animals within Q1 was 63% and 70%, respectively. Those in Q4 were classified with an accuracy of 42% by RBT and 54% by RBT+ (Table 7).

Table 3. Descriptive statistics for cow-lactations retained in the final CFI and CFI+ random forest models for the prediction of conception to first insemination between DIM 22 and 80

Variable	Total (range)
No. of cow-lactations	1,130
No. of farms	13
Production system	
All year-round calving, fully housed	10
Spring calving, grazing	3
Mean no. of cow-lactations/herd	87 (30–219)
Mean peak yield,[1] herd level (kg)	54 (39–69)
Conception rate to first insemination	
Total (%)	42
Parity 2 (%)	43
Parity 3+ (%)	41
Mean conception rate to first insemination, herd level (%)	46 (21–69)
Mean DIM at first insemination, herd level (%)	61 (54–70)

[1]Calculated using maximum single-day yield recorded in the first 100 DIM.

DISCUSSION

This is the first study to report the utility of transition cow data collected by AMS for the prediction of reproductive performance. By comparing these models with those utilizing 2 additional data sources, we also report the marginal utility provided by rumination and activity data, as well as historical cow-level data. Though the performance described is comparable with previously reported models, their current utility for the implementation of TRM is limited by poor classification accuracy within key groups. Of note within this study is the failure of the addition of auxiliary data sources to increase the accuracy of prediction over models built using AMS data alone.

The early resumption of cyclicity postcalving is key to subsequent reproductive performance. Approximately 25% of cows are reported to be acyclic by d 50 to 60 postcalving (Walsh et al., 2007; Santos et al., 2009; Pinedo et al., 2020). Animals suffering this delayed resumption of cyclicity experience decreased rates of insemination (Borchardt et al., 2021) and conception (Galvão et al., 2010) in early lactation.

The detection of estrus during the voluntary waiting period is a commonly used proxy for the assessment of ovarian cyclicity (Fricke et al., 2014; Rial et al., 2022). We chose to assess cyclicity using EOI between d 22 and 65 postcalving. Within a TRM strategy, the ability to predict the likelihood of EOI may be used to minimize the number of acyclic animals at the start of their breeding period through targeted pre-breeding screening or hormone treatment (Rhodes et al., 2003). The EOI-RBT and EOI-RBT+ models classified animals with an accuracy of 60% and 57%, respectively, demonstrating poor model discrimination. When assessed by quartile, the accuracy with which the RBT+ model classified animals least likely to record an estrus event (Q1) was moderate at 72%. However, this model failed to classify those in Q3

Table 4. Scaled variable importance for the EOI-RBT model, a random forest model for the prediction of expression of estrus from DIM 22–65 (EOI) utilizing AMS data exclusively

Variable	Variable importance score[1]
Mean concentrate intake	100
Mean milk yield	90
Mean conductivity	51
Mean refusals	26
Mean protein	0

[1]Variable importance score as calculated by the VarImp function of the CARET package (Kuhn, 2008).

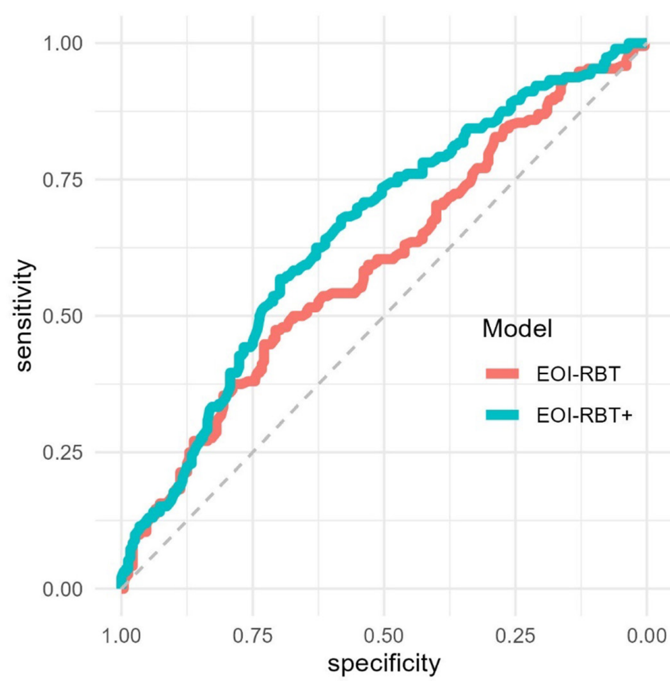

Figure 2. ROC curve for EOI-RBT and EOI-RBT+ random forest models for the predictions of expression of estrus between d 22 and 65 postcalving as evaluated on the test dataset.

Table 5. Classification performance of all random forest models built utilizing AMS data exclusively (RBT) and AMS data in conjunction with auxiliary data (RBT+) for the prediction of expression of estrus from DIM 22–65 (EOI) and conception to first insemination from DIM 22–80 (CFI) as assessed on the test dataset[1]

Item	Model			
	EOI-RBT	EOI-RBT+	CFI-RBT	CFI-RBT+
AUC-ROC	0.60	0.65	0.56	0.62
Accuracy	60	57	62	59
Sensitivity	38	15	25	25
Specificity	79	91	88	80
PPV	60	59	53	44
NPV	60	57	64	63
McNemar *P*-value	0.9		0.5	

[1]AUC-ROC = area under the receiver operator curve; PPV = positive predictive value; NPV = negative predictive value.

or Q4 with an accuracy above 50%, limiting its ability to reliably inform management decisions.

The ability to predict the likelihood of CFI may facilitate the selective breeding of animals, such as the targeted use of sexed semen in animals with a high probability of conception. We investigated the likelihood of CFI within the first 80 d postcalving. This incorporates the first 6 wk of the breeding season, assuming a 42-d voluntary waiting period, a time where the targeted use of sexed semen is commonly advocated (Butler et al., 2014). The CFI models demonstrated poor discriminative power, though the accuracy of classification for Q1 and Q2 within the CFI-RBT+ model was moderate, 70% and 72%, respectively. However, both CFI models demonstrated a lack of predictive power for Q4, limiting the confidence with which this highly fertile subgroup could be targeted with sexed semen.

The utility of transition cow data for the prediction of subsequent fertility within our dataset is limited, though the accuracy of CFI models is comparable with models previously reported (Shahinfar et al., 2014; Hempstalk et al., 2015; Barden et al., 2024). Although the potential value that TRM strategies may provide cannot be measured solely in the accuracy of their predictive models (Harris, 2017), they do serve as the foundation of their utility. For the on-farm implementation of TRM relating to EOI and CFI, the accuracy of classification within the groups at both extremes of reproductive performance is key. None of the 4 models investigated here demonstrated satisfactory accuracy across both Q1 and Q4 (Table 7). Although we believe these models have demonstrated potential for the prediction of fertility, improved classification performance will be required prior to their implementation within a TRM strategy.

Developing a more detailed representation of each animal through the integration of additional data sources is often cited as a potential means of improving model performance (Shahinfar et al., 2014; Dallago et al., 2019;

Table 6. Scaled variable importance for the EOI-RBT+ model, a random forest model for the prediction of expression of estrus from DIM 22–65 (EOI) utilizing AMS and auxiliary data

Variable	Variable importance score[1]
Mean concentrate intake	100
Variation activity	93
PV-mean milk yield	84
Mean activity	82
Mean milk yield	70
AFC	60
Mean protein	3
PV-DC[2]	0

[1]Variable importance score as calculated by the VarImp function of the CARET package (Kuhn, 2008).

[2]DIM at conception in prior lactation.

Table 7. Classification accuracy (%) per quartile of all random forest models for the prediction of expression of estrus from DIM 22–65 (EOI) and conception to first insemination from DIM 22–80 (CFI) as assessed on the test dataset

Model	Quartile[1]			
	Q1	Q2	Q3	Q4
EOI-RBT	61	55	53	60
EOI-RBT+	72	62	45	48
CFI-RBT	63	60	70	54
CFI-RBT+	70	72	49	42

[1]Quartiles based on the within farm predicted likelihood of a positive outcome. Q1 representing the least likely, Q4 the most likely.

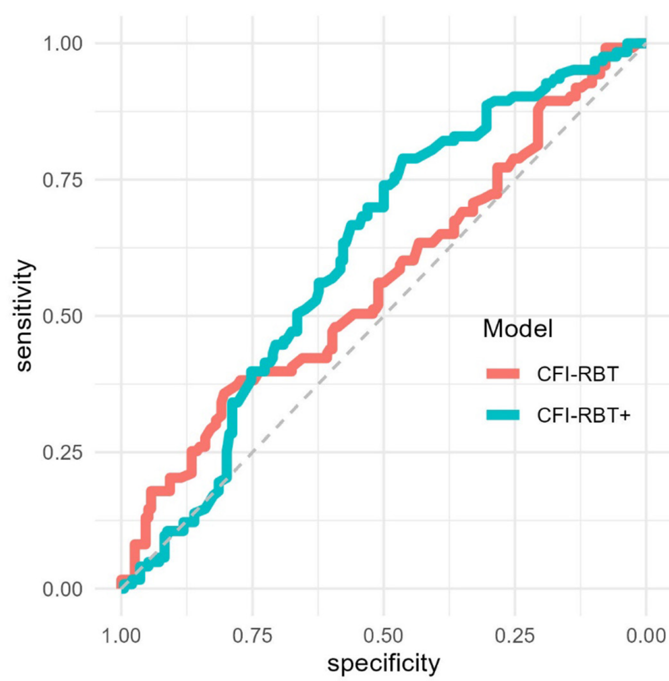

Figure 3. ROC curve for CFI-RBT and CFI-RBT+ random forest models for the prediction of conception to first insemination between d 22 and 80 postcalving as evaluated on the test dataset.

Table 8. Scaled variable importance for the CFI-RBT Model, a random forest model predicting conception to first insemination from DIM 22–80 utilizing AMS data exclusively (CFI-RBT)

Variable	Variable importance score[1]
Mean delta yield	100
Mean concentrate intake	94
Mean conductivity	69
Mean milk yield	57
Mean milk FPR[2]	43
Mean protein	11
Mean milkings	0

[1]Variable importance score as calculated by the VarImp function of the CARET package.

[2]FPR = fat-to-protein ratio.

Table 9. Scaled variable importance for the CFI-RBT+ Model, a random forest model predicting conception to first insemination from DIM 22–80 utilizing AMS and auxiliary data

Variable	Variable importance score[1]
Mean concentrate intake	100
Mean delta yield	75
PV-mean milk yield	67
Mean rumination	63
Mean conductivity	37
Mean milk FPR	37
Mean delta rumination	37
PV-yield at dry	25
PV-DO[2]	19
Mean protein	2
Mean milkings	0

[1]Variable importance score as calculated by the VarImp function of the CARET package.

[2]DIM at dry-off in prior lactation.

Ha et al., 2022). We investigated the change in performance following the addition of 2 auxiliary sources chosen based on their demonstrated ability to reflect transition health and influence fertility (Eastham et al., 2018; Stevenson et al., 2020; Borchardt et al., 2021). In the final EOI-RBT+ model, data from auxiliary sources comprised 5 of the 8 variables retained following RFE (Table 6). Similarly, in the final CFI+ model, data from auxiliary sources accounted for 5 of the 11 variables retained (Table 9). Despite this, no statistically significant difference was detected in the classification accuracy of the RBT and RBT+ models. The retention of these variables within the RBT+ models highlight the care that must be applied to feature selection to preserve model parsimony as the number of available data sources continues to expand.

Although the increasing adoption of technology on commercial dairy farms has the potential to improve the accuracy of predictive models, this must be balanced with the limitations it imposes on model development and deployment. For example, rumination and activity monitoring, although commonly employed by farms utilizing AMS, is not ubiquitous. Furthermore, as this technology is supplied by third parties, issues relating to the uniformity of sensors and software across farms can complicate data integration and increase data missingness. As seen within our study, incorporating such data sources reduces the volume of data available for model development. Furthermore, it introduces additional data requirements for farms wishing to employ such models, limiting its deployment. The integration of additional data sources must therefore be justified by substantial improvement in model performance.

All data used within this study was accessed via Lely's third-party application programming interface API. The use of data, which is widely collected and remotely accessible across Lely AMS, markedly increases the ease with which predictive models may be deployed commercially. However, the nature of this dataset imposed several limitations on this study. One such limitation is the absence of data relating to individual herd management practices, in particular the use of exogenous hormones and fixed-time AI. In the case of the CFI models, the use of such treatments has the potential to influence the probability of conception (Fricke and Wiltbank, 2022) and hence, bias our predictions. The use of herd-level fixed-time AI is not commonly practiced in the United Kingdom or Republic of Ireland, particularly in the first 80 d postcalving. However, to assess the degree to which such practices may have been applied within our dataset, insemination records were examined as previously described by Barden et al. (2024). In brief, for each herd, the proportion of inseminations on each day, of each week, were calculated. Thereafter, the binomial SD for a uniformly distributed proportion (i.e., 1/7) for all inseminations within each week was calculated. Any day for which the proportion of inseminations reported was in excess of 2 SD for the week was identified and all inseminations on that day marked as potentially fixed-time inseminations. Within the final CFI dataset, 8% of inseminations were identified as potential fixed-time inseminations. This indicates that the practice of herd-level fixed-time insemination is unlikely to have been used extensively within this dataset. However, we cannot conclusively state that insemination events analyzed were not influenced by such treatments, and our results must be interpreted in light of this limitation. A further limitation is our reliance on farm level recording of insemination events and pregnancy diagnosis. It was not possible for the research team to assess the means and consistency by which these events were recorded across farms, or the effect between-farm variance may have had on our results. As these events form key aspects of our analysis, this represents a major limitation within our study. This study is the first to demonstrate the potential utility of AMS data collected during transition to develop a generalizable predictive model for subsequent reproductive performance. However, due to the small number of herds used and the limited number of lactations assessed within each herd, these results must be interpreted with caution. Although we sought to incorporate a range of farming systems commonly found on AMS within the United Kingdom and Republic of Ireland, validation of these results utilizing a larger and more diverse dataset is required to more confidently assess the generalizability of these models.

For the implementation of TRM, parsimonious predictive models, which reliably deliver good to excellent classification accuracy in a wide range of commercial settings, are required. As the range of novel data sources expands, the investigation of their ability to improve model performance will be key to progress within this field. Recent studies have demonstrated the potential of data sources such as genomics (Rial and Giordano, 2024) and the in-line assessment of progesterone (Blavy et al., 2018) for the prediction of reproduction. Where such data sources were readily available for integration with AMS data, they may hold greater potential for predictive performance than the auxiliary data sources investigated here. However, future studies should not neglect the exploration of alternative methods for the improvement of model performance. For example, the clustering of farms based on environmental and management conditions may provide an opportunity to increase performance by tailoring models to each farm's specific circumstance (Ng et al., 2015). This approach has been employed on farms utilizing AMS to facilitate targeted management recommendations (Tremblay et al., 2016). However, its utility for the development of predictive models remains unexplored. Such methods may reduce the need for the integration of a large number of data sources and facilitate the implementation of TRM on a broader scale across AMS.

NOTES

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Nonstandard abbreviations used: AFC = age at first calving; AMS = automatic milking systems; AU = arbitrary unit; AUC-ROC = area under the receiver operator curve; CFI = conception to first insemination between 22 and 80 DIM; EOI = expression of estrus or insemination event between 22 and 65 DIM; FPR = Fat-to-protein ratio; NPV = negative predictive value; PPV = positive predictive value; PV-DC = DIM at conception in the prior lactation; PV-DO = DIM at dry-off; PV-mean milk yield = daily milk yield as averaged across the entire lactation; PV-yield at dry = mean milk yield for the last 10 d before drying off; Q = quartiles; RBT = dataset consisting of only AMS data; RBT+ = dataset consisting of AMS data in conjunction with 2 auxiliary data sources; RFE = recursive feature elimination; TRM = targeted reproductive management.

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